



KAMPALA INTERNATIONAL UNIVERSITY - WESTERN CAMPUS
DEPARTMENT OF PHARMACOLOGY AND TOXICOLOGY
BASIC PHARMACOLOGY TWO EXAM FOR BMS 2.1
DATE: 19TH December 2020 DURATION: 3 hours

Instructions

Answer ALL questions in section A using the special MCQ template provided and only two questions from section B using the booklet provided.

- Which of the histamine receptor subtypes is involved in gastric acid secretion by the parietal cells when stimulated by histamine
A. H₁
☒ B. H₂
C. H₃
D. H₄
- Drugs that block the receptor in question (1) above are therefore used in treatment of one of the following disease.
☒ A. Congestive heart failure
B. Ulcers
C. Hypertension
D. Diabetes
- One of the following is a good example of such drugs mentioned in question (2) above
☒ A. Loratidine
B. Moricizine
C. Famotidine
D. Cyclizine
- One of the following is commonly known as a physiological antagonist of histamine.
A. Acetylcholine
B. GABA
☒ C. Epinephrine
D. Glutamate
- One of the following is a histamine antagonist which inhibits the release of histamine
☒ A. Beta2 agonist
B. Beta 2 antagonist
C. Alpha 1 agonist
D. All of the above.
- Due to its effect, the agent in question (5) above is commonly used in the treatment of one of the following.
A. Hypertension
B. Diabetes
☒ C. Asthma
D. Heart failure
- Which of the following serotonin receptors is known to play a role in depression/anxiety
A. 5HT-1B
B. 5HT-1C
☒ C. 5HT-1A
D. 5HT-1D
- One of the drugs below is an agonist of the receptor in question (7) above used in the treatment of depression.
☒ A. Ranitidine
B. Buspirone
C. Sumatriptan
D. All of the above.
- Which of the following statements is CORRECT.
A. NSAIDs which inhibits both COX-1 and COX-2 are more preferred
B. NSAIDs which inhibits COX-1 selectively are more preferred
C. NSAIDs which inhibits COX-2 selectively are more preferred
D. All of the above
- NSAIDs which inhibits both COX-1 and COX-2 are contraindicated in patients suffering from one of the following conditions
A. Hypertension
B. Congestive heart failure
C. Ulcers
☒ D. Chronic kidney disease
- Which of the following statements is correct about the drug ZAFIRLUKAST
A. It is used in the treatment of diabetes by stimulating the release of insulin

- ☒ B. It is used in the treatment of asthma by inhibiting the enzyme 5-lipoxygenase which prevents the formation of thromboxane A₂
- ☐ C. It is used in the treatment of asthma by inhibiting the enzyme 5-lipoxygenase which prevents formation of leukotrienes.
- ☒ D. It is an antagonist of the cysteinyl leukotrienes receptors used in the treatment of asthma.
12. Which of the following is a serotonin receptor that is known to mediate nausea and vomiting.
- A. 5HT-2
- B. 5HT-1A
- C. 5HT-1D
- ☒ D. 5HT-3
13. Which of the following is correct about the drug CAPTOPRIL
- A. It is an angiotensin II receptor blocker used in the treatment of hypertension.
- B. It is an angiotensin II receptor blocker used in treatment of hypotension
- ☒ C. It is an angiotensin converting enzyme inhibitor used in the treatment of hypertension.
- D. All of the above
14. One of the options below contains an autacoid whose levels are increased by the action of the drug in question (13) above due to inhibition of its degradation.
- ☒ A. Angiotensin I
- B. Bradykinin
- C. Cholecystokinin
- D. All of the above
15. Which of the following is a prominent adverse effect caused due to the accumulation of the autacoid mentioned in question (14) above when CAPTOPRIL is used.
- A. Liver failure
- ☒ B. Dry cough
- C. Heart failure
- D. None of the above
16. Which of the following is correct about the drug CANDESARTAN
- A. It is a bradykinin antagonist used in the treatment of constipation.
- ☒ B. It is an angiotensin II receptor antagonist used in the treatment of hypertension
- C. It is an angiotensin II receptor antagonist used in the treatment of hypotension
- D. It causes vasoconstriction.
17. Hyperprolactinemia can be treated by the use of one of the drugs below
- A. Amlodipine
- ☒ B. Bromocriptine
- C. Serotonin
- D. Epinephrine
18. Which of the following is true about the drug ZILEUTON.
- A. It is used in the treatment of diabetes by stimulating the release of insulin
- B. It is used in the treatment of asthma by inhibiting the enzyme 5-lipoxygenase which prevents the formation of thromboxane A₂
- ☒ C. It is used in the treatment of asthma by inhibiting the enzyme 5-lipoxygenase which prevents formation of leukotrienes.
- D. It is an antagonist of the cysteinyl leukotrienes receptors used in the treatment of asthma.
19. Which of the following is a prostaglandin E₁ (PGE₁) analogue used in treatment of peptic ulcers disease.
- A. Dinoprostone
- ☒ B. Misoprostol
- C. Ranitidine
- D. Carbergoline
20. One of the following is a prostaglandin E₁ (PGE₁) analogue used in the treatment of erectile dysfunction.

- B. It produces lower cure rates in intestinal helminthiasis than mebendazole or albendazole
 C. It needs pretreatment fasting and post treatment purgative
 D. It is not active against roundworm and hookworm
31. One of the following anthelmintic has been found to be safe during pregnancy:
 A. Thiabendazole
 B. Piperazine
☒ C. Albendazole
 D. Pyrantel pamoate
32. The drug below is most recommended for treatment of *Schistosoma haematobium*, *Schistosoma mansoni* and other trematode infections but should be avoided in ocular cysticercosis
 A. Albendazole
 B. Mebendazole
☒ C. Praziquantel
 D. Thiabendazole
33. Select the condition for which 3 days treatment with pyrantel pamoate is recommended in place of single dose therapy for others:
☒ A. Ascariasis
 B. Ancylostomiasis
 C. Necatoriasis
 D. Enterobiasis
34. The following drug is the drug of choice in filariasis caused by infection with *Wuchereria bancrofti* and *Brugia malayi* but should be avoided in patients with severe Mazzotti reactions with onchocerciasis.
☒ A. Diethylcarbamazine
 B. Ivermectin
 C. Mebendazole
 D. Praziquantel
35. Which of the drug/drugs below is effective in filariasis?
 A. Ivermectin
 B. Albendazole
☒ C. Diethyl carbamazine citrate
☒ D. All of the above
36. Diethyl carbamazine citrate has one of the following action in filariasis:
 A. Kills circulating microfilariae
 B. Kills microfilariae present in nodules and serous fluids
 C. Promotes phagocytosis of circulating microfilariae
☒ D. Rapidly kills adult filarial worms and stops production of microfilariae
37. Select the drug that is used orally to treat scabies:
 A. Permethrin
☒ B. Ivermectin
 C. Praziquantel
 D. Crotamiton
38. Laboratory examination of a 20 yr-old migrant from the neighbouring Democratic Republic of Congo found him to be parasitized by both schistosomes and *T. Solium*—the pork tapeworm. Which of the following anthelmintic drugs would be effective for both infestations?
 A. Albendazole.
 B. Ivermectin.
 C. Mebendazole.
☒ D. Praziquantel.
39. Which anthelmintic drug acts through a specific glutamate gated Cl^- ion channel found only in nematodes:
☒ A. Ivermectin
 B. Niclosamide
 C. Pyrantel pamoate
 D. Praziquantel
40. A saline purgative is recommended for the following:
 A. Albendazole for whipworm infestation (Trichuriasis)
 B. Mebendazole for roundworm infestation
☒ C. Niclosamide for tapeworm infestation

- D. Pyrantel pamoate for hookworm infestation
41. The drug of choice for neurocysticercosis is:
 A. Praziquantel
 B. Niclosamide
 C. Ivermectin
D. Albendazole
42. Bacterial resistance is a common cause of treatment failure in patients suffering from bacterial infections. How would we overcome this problem?
 1. By prescribing broadspectrum antibacterials to target both Gram positive and gram negative bacteria.
 2. By doing culture and sensitivity for all patients on antibacterials
 3. By using narrow spectrum antibacterials.
 4. By indiscriminate use of antibacterials
 A. Only 1, 2 and 3 are correct
 B. Only 1 and 4 are correct
 C. Only 2 and 3 are correct
 D. All the above
43. Which of the following can help reduce the prevalence of surgical site infections among post caesarean mothers?
 1. Reducing the number of caesarean sections done
 2. Having a good antimicrobial stewardship program
 3. Making sure all patients receive prophylactic antibiotics atleast 1 hour before surgery
 4. Ensuring wound surveillance is done
 A. Only 1, 3 and 4 are correct
 B. Only 2 and 4 are correct
 C. Only 2, 4 and 3 are correct
 D. All the above
44. Which of the following is the best quick screening diagnostic test for TB in an HIV/AIDS patient?
 A. Gene Expert test
 B. Mid Upper Arm Circumference
 C. LAM test for urine
 D. Gram staining
45. All of the following are aminoglycosides used in treatment of Tuberculosis as per WHO guidelines except
 1. Amikacin 2. Kanamycin 3. Streptomycin 4. Capreomycin 5. Gentamycin
 A. 1, 2 and 3
 B. 3, 4 and 5
 C. 3, 4 and 5
 D. only 4
46. You are employed in a TB unit at a health centre IV and a patient develops MDR-TB, as part of the health care team, what would be your recommendation for this patient?
 1. Refer the patient to a TB specialist
 2. Recommend pretomanid and bedaquiline
 3. Recommend culture and sensitivity
 4. Recommend amikacin, capreomycin and Kanamycin
 A. Only 1 and 2
 B. Only 1 and 3
 C. Only 1, 2 and 3
 D. Only 3 is correct
47. What is the maximum duration for secondary prophylaxis of cryptococcal meningitis using fluconazole?
 A. One year
 B. For life
 C. Six months
 D. As long as CD4 is above 100 cells/mm³
48. What is the mechanism of amphotericin B induced nephrotoxicity?
 A. It causes ischemic Tubular injury
 B. It reduces renal blood flow
 C. It causes glomerular injury
 D. It causes inflammatory tubular injury

1.5m
10ms

49. In order to minimize amphotericin-B induced nephrotoxicity, what would be the maximum recommended cumulative dose of amphotericin-B?

1. A. 2-3 g C. not more than 5g
2. B. 1-2 g D. 3-4 g

50. Which of the following is not true about the pharmacological properties of ketoconazole?

- A. Only administered orally, requires gastric acid for dissolution.
B. Effective in histoplasmosis in lung, bone, skin and soft tissues.
C. The drug has a good penetration into the CSF (cerebral spinal fluid).
D. The drug may cause gynaecomastia

51. When loop diuretics are given together with amphoterin B, which of the following adverse effects is expected to be additive?

1. Chills and fever 2. Nephrotocity 3. Ototoxicity 4-
Hypokalemia

- A. Only 1 and 2 are correct C. Only 2 and 3 are correct
B. 1, 2 and 3 are correct D. 2, 3 and 4 are correct

52. Mr Okalodhiang while working in the Experimental Pharmacology Lab. stumbled and spilled a tube of infected mouse tissue on the floor which he got in contact with. He is most probably at risk of..... Infection.

- A. Coccidioidomycosis ✓ Babesiosis
B. Tularemia D. None of the above

53. Ms. Nyeno Loyo Ogowang, during Toxicity testing with mice got involved with an accident where she sustained an intra-nasal inoculation. This student at risk of developing.....Desease

- A. Brucellosis C. Coxsackie virus infection
B. Rabbits D. None of the above

54. You have been called upon to advise year one medical students about safety while performing their experiments in the Pharmacology lab. Which BSL are they supposed to use?

- A. BSL 1&2 C. BSL 3 & 4
B. BSL 2&3 D. None of these

55. In Biological & Clinic settings, the main source of contamination is?

- A. Animal bites C. Radio active elements
B. Hazardous chemicals D. None of these

56. Recently, a published report indicated that Kampala city is among the 10 top most polluted cities in the world. Which branch of Toxicology deals with this type of information?

- A. Occupational toxicology C. Biological toxicology
B. Ecotoxicology D. Environmental toxicology

57. What is the duration of Special Toxicity testing?

- A. 14 days C. 90 days
B. 42 day D. None of these are

58. Which of the following statements is not true?

- A. Since DFT is stored in the fat, rapid mobilization of fat can rapidly increase plasma conc of the toxicant
B. Lead is usually stored in the bone with dissastirous results
C. Placental barrier cannot limit the distribution of the drug to the fetus

59. Kaakan - a boy of 7 years tries mimicking by mixing 1 tea spoonful of the paraquat in water. He took 1 teaspoonful to 2 gallons water. The resultant solution is..... ?

A. 1000ppm

B. 200ppm

C. 500ppm

D. 100000ppm

60. Phagiphaia, a medical student at KIU was doing his research about contamination of some herbal products from Ishaka market by aflatoxins. This student found 1 micrograms of this toxic ant in 1 kg. This can be expressed as?

A. 1PPB

B. 10PPB

C. 100PPB

D. 1000PPB

61. Which of the following statement is true about NOEL?

A. High dose of a poison that produces noticeable toxic effects

B. High dose of a poison that produces no noticeable toxic effects

C. Low dose of a poison that produces no noticeable toxic effects

D. Low dose of a poison that produces noticeable toxic effects

62. Madam Kinominom, who is also an HIV patient is 11 weeks pregnant. Should she be started on malaria Iptp or not?

A. Yes, to protect her from malaria

B. No, because she is on cotrimoxazole

C. No, this could pose risk of abnormality

D. None of the above

63. At 20 weeks of pregnancy, madam Kinominom visits a nearby village clinic because she is not feeling well. The owner of the clinic gives her Folic acid 5mg/day with pyrimethamine and sulphadoxin for her IPTp. On reaching home, her sister who is an intern at KIUTH asks to first cross check with her notes before drug admin. What's the problem?

A. Intern is wrong: this is the right time to take both drugs

B. Nurse has given her the IPTp but with low dose of Folic acid

C. It is a wrong time for the IPTp to be given with Folic Acid

D. None of the above

64. Your white friend from the USA visits you in Uganda for the very first time from the UK. What prophylactic antimalarial would you give your visitor?

A. Mefloquine

B. Quinine

C. Halofantrine

D. Artesunate

65. Mr Oopoloo is on rifampicin as one of the drugs for treating his TB. He is also diagnosed with malaria. He is given quinine and paracetamol tablets to take besides his rifampicin. What is your advice?

A. This was done in error as quinine is Contra indicated in this case

B. This combination will lead to a reduction in rifampicin plasma level & reduce the effectiveness of rifampicin

C. This combination will lead to a reduction in the effectiveness of quinine against malaria

D. None of the above

66. Which of the following is Not True regarding 8- aminoquinolines currently used as antimalarials?

A. Have broad spectrum activity

B. Have excellent efficacy

C. Present with hematological toxicities

D. None of these

67. Which class of the below enlisted drugs has found use in the treatment of Pneumocystis pneumonia?
- | | |
|----------------------|------------------------|
| A. 8-Aminoquinolines | C. Chinchona alkaloids |
| B. Methyl Alcohols | D. Artemisinins |
68. Suramin is the commonly used drug for the treatment of?
- | | |
|------------------------------|----------------------------|
| A. T. b. gambiense infection | C. African trypanosomiasis |
| B. Leishmaniasis | D. Babesiosis |
69. The drug of choice for the treatment of T. trichuria is?
- | | |
|------------------|-----------------|
| A. Mebendazole | C. Praziquantel |
| B. Thiabendazole | D. Pentamidine |
70. The commonly recommended treatment for amoebic dysentery is?
- A. Metronidazole followed by diloxanide
 B. Mebendazole followed by chloroquine
 C. Albendazole followed by Tinidazole
 D. Thiabendazole followed by Acetyl Salicylic Acid
71. Which of the following agents have potential carcinogenic effects, and may therefore be important targets in anticancer interventions
- A. Alcohol, viruses, and Tobacco
 B. Tobacco, High fibre diet, and Viruses
 C. Ultraviolet light, Tobacco and Asbestos
 D. All the above,
 E. A and C
72. From the list below, select the antiretroviral agent that inhibit the reverse transcriptase enzyme
- | | |
|---------------|-----------------|
| A. Saquinavir | D. Enfuvirtide |
| B. Abacavir | E. Dolutegravir |
| C. Maraviroc | |
73. Which of the following drugs inhibits the neuraminidase enzyme used by influenza
- | | |
|----------------|----------------|
| A. Rimantadine | C. Oseltamivir |
| B. Ganciclovir | D. Enfuvirtide |
74. Select the most appropriate targeted phase of cell division for Cyclophosphamide
- | | |
|----------------------|---------------------|
| A. Mitosis | D. B and C |
| B. The resting phase | E. All of the above |
| C. Synthesis Phase | |
75. Which of the following antiretroviral agents has the longest plasma half-life, concentrates intracellularly, and is also currently recommended for first line treatment for HIV in Uganda?
- | | |
|------------------|-----------------|
| A. Abacavir | D. Zidovudine |
| B. Emtricitabine | E. Dolutegravir |
| C. Lopinavir | |
76. Which of the following antiretroviral agents has proven inhibitory action against corona viruses and malaria parasites?
- | | |
|------------------|-----------------|
| A. Abacavir | D. Zidovudine |
| B. Emtricitabine | E. Dolutegravir |
| C. Lopinavir | |
77. Select the best treatment option for hepatitis B

- A. Boceprevir
- B. Ritonavir
- C. Tenofovir

- D. Pegylated Interferon
- E. Efavirenz

78. Which of the following are general mechanisms by which organisms may develop resistance to antimicrobial agents

- A. By developing alternative pathways
- B. Through gene mutation
- C. By inhibiting intracellular accumulation of the drug
- D. All the above
- E. B and C, only

79. Which of the following agents would treat both Hepatitis B and C viruses

- A. Ritonavir
- B. Boceprevir
- C. Abacavir
- D. Interferon alfa
- E. Remdesivir

80. Which of the following drugs only inhibits DNA synthesis in cancer cells

- A. Melphalan
- B. Mercaptopurine
- C. Methyl prednisolone
- D. Mitomycin
- E. Mitoxantrone

For questions 81 to 85, regarding sars cov-2 virus, indicate with 'T' for correct statements, or 'F' for wrong statements.

81. Remdesivir is a protease inhibitor approved for SARS -COV-2 -----

82. Corticosteroids may be administered to counteract the immune reaction but not to kill the virus ----F----

83. Chloroquine and hydroxychloroquine are potent inhibitors of SARS-COV-2 viral entry-T

84. The currently approved BNT162b2 is an mRNA-based vaccine, with efficacy of 95% ---F

85. The currently approved BNT162b2 is a DNA-based vaccine, with efficacy of 95% --- F

For questions 86 to 90, regarding dolutegravir, indicate with 'T' for correct statements, or 'F' for wrong statements (1 mark each)

86. Dolutegravir is a protease inhibitor with plasma half-life of 2-3 hours F

87. Dolutegravir is primarily metabolized by Cytochrome P450 enzymes F

88. Dolutegravir can be used in first line regimens for children above 20kg T

89. Dolutegravir lacks significant drug-drug interactions with antimalarial drugs T

90. The half-life of dolutegravir is longer than that of cabotegravir F

From 91-92 choose the odd man out which is odd by being either the only one correct or wrong.

91. Regarding chemotherapeutic agents

- a) Antibiotics agents are any chemical substance that can either kill or inhibit the growth of micro-organisms regardless of its origin
- b) Antimicrobial agents are chemical substances produced by natural metabolic processes that kill or inhibit the growth of micro-organisms
- c) Chloramphenicol is an antibiotic but not an antimicrobial agent
- d) Not all antimicrobial agents are antibiotics

92. Regarding emergence of resistance

- a) Indiscriminate use of antimicrobials increases resistance
- b) Use of drug combinations reduces resistance
- c) Unnecessary use of antimicrobials promotes resistance
- d) Adherence to treatment increases resistance

For question 93 to 97, match the antimicrobial agents given below with their appropriate prophylactic uses

- A. chloroquine
- B. minocycline
- C. clotrimazole

- D. cefotaxime
- E. penicillins

- B 93. Patients with meningococcal meningitis *cefotaxime*
- B 94. Dental procedures in people at risk of endocarditis *penicillins*
- C 95. Children with repeated urinary tract infections *penicillins*
- D 96. Perioperative prophylaxis in certain types of surgery where the risk of infection is high e.g. colorectal or the consequences of infection are life threatening e.g. open heart *minocycline*
- A 97. Prophylaxis of malaria *chloroquine*

From question 98-100, match the compounds below with the mechanisms of action provided.

- A. Sulfamethoxazole
- B. Vancomycin
- C. Pyrimethamine

- D. Bacitracin
- E. Ceftriaxone

- 98. Blocks transportation of NAG and NAM subunits from cytoplasm to the growing peptidoglycan *Bacitracin*
- A 99. Dihydropteroate synthetase inhibition *antibacterial sulfonamides*
- B 100. Hinders peptidoglycan elongation by interfering with particular bridges that link NAM subunits in many Gram-positives *Bacitracin Vancomycin*

SECTION B (ESSAY)

Answer only two questions from this part (50 marks)

1. You have been employed as the Head of department at the Surgical ward of KIU-Teaching Hospital and there is a high rate of surgical site infections among post-operative patients. Using your knowledge on the principles of Chemotherapy, use a table elucidate how you can mitigate this problem. (25 marks)

2(a). A 26 year old pregnant woman presents to KIU teaching hospital complaining of a 1-week history of dysuria, Lower abdominal pain, fever and foul-smelling vaginal discharge. A first-catch urine specimen is obtained and tested positive for gonococcal infection. Describe in details the factors you will considered before choosing a suitable chemotherapeutic agent for this patient.

(b) Basing on the case provided above in question two, Name the drug of choice that you would use to manage the patient

(c). Give a reason as to why you have chosen the drug in (a) above

- (d) Name the class and subclass to which the drug you have chosen belongs to
- (e) State the mechanism of action for your drug of choice
- (f) List at least 4 side effects of the above drug
- (g) Name at least 4 antibacterial agents you would not administer to this patient

3(a)(i) Write down two therapeutic uses of the H₁ antihistamines with two examples of drugs used in each condition mentioned. (6 marks)

ii) State two adverse effects each for the H₁ antihistamines when used for each of the cases mentioned above (4 marks)

iii) Name ONE therapeutic use including the mechanism of action of the H₂ receptor antagonist (H₂ anti-histamines) with two examples of drugs used in the condition mentioned. (5 marks)

b) In a tabular form, mention three serotonin agonists, their respective clinical applications (therapeutic use). (6 marks)

Example:

S/N	Serotonin agonists	Clinical applications
1		
2		
3		

c) In a tabular form as given the example in (b) above, mention two serotonin antagonists and their clinical applications. (4 marks).

4(a) Describe the concept of selective toxicity in antiviral therapy, and give two examples of antiviral agents specifying how they achieve selective toxicity (5 marks)

(b) Describe the role of gene mutations in cancer development and give two examples of anticancer agents which target mutated genes (5 marks)

(c) Write brief notes on Tamoxifen, including its mechanism of action (2marks), clinical application (1 mark) and two unwanted effects (2 marks)

(d) List three antiretroviral drugs (2 marks) used in post exposure prophylaxis (PEP), and explain their pharmacokinetic properties that facilitate this role (3 marks)

- (d) Name the class and subclass to which the drug you have chosen belongs to
- (e) State the mechanism of action for your drug of choice
- (f) List at least 4 side effects of the above drug
- (g) Name at least 4 antibacterial agents you would not administer to this patient

3(a)(i) Write down two therapeutic uses of the H₁ antihistamines with two examples of drugs used in each condition mentioned. (6 marks)

ii) State two adverse effects each for the H₁ antihistamines when used for each of the cases mentioned above (4 marks)

iii) Name ONE therapeutic use including the mechanism of action of the H₂ receptor antagonist (H₂ anti-histamines) with two examples of drugs used in the condition mentioned. (5 marks)

b) In a tabular form, mention three serotonin agonists, their respective clinical applications (therapeutic use). (6 marks)

Example:

S/N	Serotonin agonists	Clinical applications
1		
2		
3		

c) In a tabular form as given the example in (b) above, mention two serotonin antagonists and their clinical applications. (4 marks).

4(a) Describe the concept of selective toxicity in antiviral therapy, and give two examples of antiviral agents specifying how they achieve selective toxicity (5 marks)

(b) Describe the role of gene mutations in cancer development and give two examples of anticancer agents which target mutated genes (5 marks)

(c) Write brief notes on Tamoxifen, including its mechanism of action (2marks), clinical application (1 mark) and two unwanted effects (2 marks)

(d) List three antiretroviral drugs (2 marks) used in post exposure prophylaxis (PEP), and explain their pharmacokinetic properties that facilitate this role (3 marks)